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United States Patent	12416865
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Wolke; Conrad et al.

Optical system, in particular for microlithography, and method for operating an optical system

Abstract

An optical system, in particular for microlithography, comprises a laser light source for generating a multiplicity of light pulses, and a control unit configured to control the laser light source in such a way that, for a light pulse sequence generated by the laser light source, the time period between respectively successive light pulses varies across the light pulse sequence. A method comprises operating the optical system.

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Appl. No.: 18/186396

Filed: March 20, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230314959 A1	Oct. 05, 2023

Foreign Application Priority Data

DE	102022107633.4	Mar. 30, 2022
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Publication Classification

Int. Cl.: G03F7/20 (20060101); G03F7/00 (20060101)

U.S. Cl.:

CPC **G03F7/70583** (20130101); **G03F7/70025** (20130101); **G03F7/70041** (20130101);

Field of Classification Search

CPC: G03F (7/70583); G03F (7/70025); G03F (7/70041); G03F (7/702); G03F (7/70175);
 G03F (7/7055); G02B (26/06); G02B (27/48); H01S (3/0057); H01S (3/225)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims benefit under 35 U.S.C. § 119 to German Application No. 10 2022 107 633.4, filed Mar. 30, 2022. The contents of this application is hereby incorporated by reference in its entirety.

FIELD

(2) The disclosure relates to an optical system, in particular for microlithography, and to a method for operating an optical system. The disclosure is applicable in particular in a laser light source for use in a microlithographic projection exposure apparatus.

BACKGROUND

(3) Microlithography is used to produce microstructured electronic components. The microlithography process is carried out in what is known as a projection exposure apparatus, which

has an illumination device and a projection lens. The image of a mask (=reticle) illuminated via the illumination device is projected in this case via the projection lens onto a substrate (for example a silicon wafer) that is coated with a light-sensitive layer (photoresist) and arranged in the image plane of the projection lens in order to transfer the mask structure to the light-sensitive coating of the substrate.

(4) In a projection exposure apparatus designed for operation in the DUV range (e.g. at operating wavelengths of less than 250 nm, in particular less than 200 nm), use is typically made of laser light sources in the form of excimer lasers, such krypton fluoride excimer lasers at an operating wavelength of 248 nm or argon fluoride excimer lasers at an operating wavelength of 193 nm.

(5) Existing challenges in the course of the development of projection exposure apparatuses concern, firstly, increasing desired accuracy properties for microlithographically producing ever smaller structures and positioning them on the wafer and, secondly, also an increase in the throughput achieved with the respective projection exposure apparatus, this increase being desirable from the standpoint of cost considerations.

(6) An effect that can occur with regard to the desired accuracy properties to be satisfied inter alia during operation of the laser light sources mentioned above is the occurrence of what are known as speckle patterns, which are attributable to the spatial and temporal coherence of the generated light. Such speckle patterns result inter alia in a reduced overlay accuracy of structures produced in different lithography steps and thus ultimately in an impairment of the optical performance of the projection exposure apparatus.

(7) As far as the abovementioned increase in the throughput achieved with the respective projection exposure apparatus is concerned, an issue is that limits are imposed on an increase in the pulse energy and/or an increase in the repetition rate, each of these increases being desirable for increasing the output power of the laser light source. In this regard, an increase in the pulse energy can lead to a higher radiation loading of optical components within the laser light source and also a greater stress for electronic assemblies. One approach known in this respect involves pulses (having a pulse length of e.g. 20 ns) generated by the laser light source being stretched to a longer time duration of e.g. (100-450) ns by the use of optical pulse stretchers and degradation of downstream optical components thus being reduced accordingly. However, limits are imposed on such stretching of the pulse length owing to the structural space restrictions that exist for the optical pulse stretcher as well, and the decreasing output power as the number of reflections increases. Moreover, there is still the risk of degradation of those optical components which are situated upstream of the optical pulse stretcher in relation to the optical beam path.

(8) On the other hand, the increase in the repetition rate and thus in the frequency of electrical discharges within the resonator of the laser light source, the increase likewise being conceivable for increasing the output power of the laser light source, can lead to acoustic resonances and ultimately to the spectral bandwidth rising above an acceptable measure.

(9) Reference is made, purely by way of example, to WO 2018/132198 A1, U.S. Pat. Nos. 10,451,890 B2, 11,054,665 B2, 7,782,922 B2, and the publications by H. Miyamoto et al.: "Next generation ArF Laser technologies for multiple-patterning immersion lithography supporting leading edge processes", Optical Microlithography XXXI, Proc. of SPIE Vol. 10587 (2018), 1058710-1 to 1058710-8, and K. Kakizaki et al.: "Ultra-high-repetition-rate ArF excimer laser with long pulse duration for 193-nm lithography", Optical Microlithography XIV, Proc. of SPIE Vol. 4346 (2001), pages 1210-1218.

SUMMARY

(10) The present disclosure seeks to provide an optical system, in particular for microlithography, comprising a laser light source and a method for operating an optical system which make it possible to reduce the occurrence of speckle patterns and also to increase the output power of the laser light source while at least partly avoiding the problems described above.

(11) According to the disclosure, an optical system, in particular for microlithography, comprises a

laser light source for generating a multiplicity of light pulses, and a control unit configured to control the laser light source in such a way that, for a light pulse sequence generated by the laser light source, the time period existing between respectively successive light pulses varies across the light pulse sequence.

(12) A concept underlying the disclosure, in particular, in an optical system comprising a laser light source for generating a multiplicity of light pulses, is for the time period existing between two successive light pulses within a pulse succession not to be chosen to be constant in a conventional way, but rather to be varied. In this respect, the disclosure includes, in particular, a deliberate detuning of the laser light source (in the sense of a departure from an operating state with a one-off “optimized” temporal interval between successive light pulses) with the aim firstly of achieving a reduction of the speckle contrast (and thus an improvement in the overlay performance of the optical system, this being crucially determined by the speckle contrast) and secondly of also making it possible to increase the repetition rate of the laser light source (and thus to increase the throughput during operation of the optical system) while avoiding an excitation of undesired (chamber) resonances in the laser light source. In particular, the variation according to the disclosure of the time period existing between two successive light pulses within a pulse succession or light pulse sequence has the effect that the light field is different for each of the relevant light pulses, with the consequence that the temporal to and spatial coherence that is crucial for the speckle contrast is destroyed.

(13) In accordance with one embodiment, the control unit is configured to control at least one actuator for manipulating the position of at least one optical component of the laser light source.

(14) In accordance with one embodiment, the optical system comprises an optical pulse stretcher having a plurality of mirrors. In this case, the optical component controlled at the actuator can be, in particular, a mirror of the optical pulse stretcher.

(15) In accordance with one embodiment, the control unit is configured to variably adjust a temporal delay of a trigger signal generated for triggering an energy feed into a laser medium of the laser light source. This embodiment can allow for undesirable features arising of acoustic resonances as described in the introduction to be efficiently prevented before the light generated by the laser light source reaches further components such as an optical pulse stretcher, for instance. The laser medium can be the laser medium of a master oscillator of the laser light source and/or the laser medium of one or more power amplifying stages (“power amplifier”) of the laser light source.

(16) In accordance with one embodiment, the optical system furthermore comprises a first measuring unit for measuring a variable that is characteristic of a speckle contrast of the light generated by the laser light source. In this case, the control unit can be configured in particular to vary the temporal pulse succession of the light pulses depending on output signals of the first measuring unit.

(17) In accordance with one embodiment, the optical system furthermore comprises a second measuring unit for measuring a variable that is characteristic of the bandwidth of the light generated by the laser light source. The control unit can in particular be configured here to vary the temporal pulse succession of the light pulses depending on output signals of the second measuring unit.

(18) In accordance with one embodiment, the control unit comprises a random number generator for the random variation of the time period existing between each two successive light pulses.

(19) In accordance with one embodiment, the control unit is configured to continuously increase or to continuously reduce the time period existing between each two successive light pulses between a predefined lower limit value and a predefined upper limit value.

(20) In accordance with one embodiment, the laser light source is designed for generating the light pulses with a repetition rate of at least 7 kHz.

(21) In accordance with one embodiment, the optical system is designed for an operating wavelength of less than 250 nm, in particular for an operating wavelength of less than 200 nm.

- (22) The disclosure furthermore also relates to a method for operating an optical system, in particular for microlithography, wherein the optical system comprises a laser light source for generating a multiplicity of light pulses, and wherein the laser light source is controlled in such a way that for a light pulse sequence generated by the laser light source, the time period existing between respectively successive light pulses varies across the light pulse sequence.
- (23) In accordance with one embodiment, this variation is adjusted on the basis of a measurement of a variable that is characteristic of a speckle contrast of the light generated by the laser light source.
- (24) In accordance with one embodiment, the variation is adjusted on the basis of a measurement of the respective bandwidth for at least one light pulse generated by the laser light source.
- (25) In accordance with one embodiment, the variation is adjusted in such a way that the average bandwidth of the light pulses generated by the laser light source is reduced in comparison with the average bandwidth that results without this variation.
- (26) In accordance with one embodiment, the variation is adjusted in such a way that the average bandwidth difference between successive light pulses is reduced in comparison with the average bandwidth difference that results without this variation.
- (27) In accordance with one embodiment, the variation is at least partly effected by manipulating the position of at least one optical component of the laser light source, in particular the position of at least one mirror of an optical pulse stretcher.
- (28) In accordance with one embodiment, the variation is at least partly effected by variably adjusting a temporal delay of a trigger signal generated for triggering an energy feed into a laser medium of the laser light source.
- (29) Further embodiments of the disclosure are evident from the description and the dependent claims.
- (30) The disclosure will be explained in greater detail below on the basis of an exemplary embodiment that is illustrated in the accompanying figures.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In the figures:
- (2) FIG. 1 shows a diagram for elucidating a variation according to the disclosure of the time period existing between successive light pulses in an optical system in accordance with a first embodiment;
- (3) FIG. 2 shows a diagram for elucidating a variation according to the disclosure of the time period existing between successive light pulses in an optical system in accordance with a second embodiment;
- (4) FIG. 3 shows diagrams for elucidating a variation according to the disclosure of the time period existing between successive light pulses in an optical system in accordance with a third embodiment;
- (5) FIG. 4 shows a schematic illustration for elucidating the possible set-up of an optical system according to the disclosure;
- (6) FIG. 5 shows a block diagram for elucidating further the possible functioning of an optical system according to the disclosure; and
- (7) FIG. 6 shows a schematic illustration for elucidating the possible set-up of a microlithographic projection exposure apparatus designed for operation in the DUV.

DETAILED DESCRIPTION

- (8) Embodiments of the present disclosure are explained below with reference to the diagrams in FIGS. 1-3 and also the schematic illustrations in FIGS. 4-5.

(9) What these embodiments have in common is that in an optical system comprising a laser light source for generating a multiplicity of light pulses, the time period existing between two successive light pulses within a pulse succession or a light pulse sequence is not to be chosen to be constant in a conventional way, but rather is varied, with the aim firstly of achieving a reduction of the speckle contrast (and thus an improvement in the overlay performance of the optical system) and secondly of making it possible to increase the repetition rate of the laser light source (and thus to increase the throughput during operation of the optical system) while at the same time avoiding an excitation of undesired (chamber) resonances.

(10) Referring firstly to FIG. 1, in a first embodiment, the time period between two successive light pulses emitted by the laser light source can be varied randomly across a light pulse sequence. In this case, a lower limit value and an upper limit value can be predefined, in particular, within which this random variation can be carried out. The vertical dashed lines represent equidistant lines for illustration.

(11) In a second embodiment in accordance with FIG. 2, the time period between two successive light pulses emitted by the laser light source can also be continuously increased or continuously reduced over a light pulse sequence, in which case once again a lower limit value and an upper limit value can be predefined.

(12) In a third embodiment in accordance with FIG. 3, a pulse succession suitable for avoiding undesired chamber resonances can also be determined in a targeted manner and then be predefined for the further operation of the laser light source.

(13) In accordance with FIG. 3, a currently generated pulse succession **310** (as temporal profile of the laser output power) is firstly determined here for a concrete configuration of the laser light source. An amplitude spectrum **320** is then determined for this pulse succession **310** in accordance with a fast Fourier transform (FFT). Furthermore, the resulting laser bandwidth of the laser light source as a function of the laser repetition rate is determined, an exemplary profile likewise being illustrated and designated by “**330**” in FIG. 3. In this profile **330** of the laser bandwidth, occurring resonances and the associated frequency bands (designated by way of example by “**331**”, “**332**” in FIG. 3) are then identified. On the basis thereof, an “optimized” amplitude spectrum **320** suitable for eliminating these resonances **331**, **332** is determined in accordance with a fast Fourier transform (FFT), the amplitude spectrum being designated by “**340**”. The resulting laser bandwidth of the laser light source as a function of the laser repetition rate is designated by “**350**”. From this a pulse succession **360** modified in a targeted manner (corresponding to the temporal profile of the laser output power) is then determined, wherein in the present scenario in accordance with FIG. 3 and merely by way of example the temporal position of a light pulse is shifted from “**361**” to “**362**”.

(14) FIG. 4 shows, in a merely schematic illustration, the set-up possible in principle of a gas discharge laser system in the form of an excimer laser in which the present disclosure can be realized.

(15) The gas discharge laser system in accordance with FIG. 4 comprises, in particular, a seed laser **410**, a relay optical unit **420**, an amplifying stage **430** and a laser output subsystem **440**.

(16) The seed laser **410** comprises, in particular, a linewidth narrowing module **411**, a master oscillator chamber (MO=master oscillator chamber) **412**, a master oscillator output coupling unit (MO OC=“master oscillator output coupler”) **413** and a line centre analysis module **414**.

(17) The relay optical unit **420** has the effect, inter alia, of adapting or aligning the output signal of the seed laser **410** with respect to the amplifying stage **430** and can comprise, in particular, a beam expanding unit with a suitable prism arrangement and also a suitable optical retardation path.

(18) The amplifying stage **430** comprises, in particular, a chamber **432** having the laser medium, a beam reversing module **431** and a unit **433** for coupling out part of the laser beam.

(19) The laser output subsystem **440** comprises, in particular, a bandwidth analysis module **441**, an optical pulse stretcher **442** and a further module **443**. The optical pulse stretcher **442** serves to stretch the pulse length of the light pulses generated by the laser light source to a longer time

duration of the order of magnitude of (100-450) ns, for example, by deflecting the electromagnetic radiation by way of circulation paths, in order to avoid degradation of downstream optical components. The further module **443** serves firstly for qualifying the output laser radiation (e.g. with regard to degree of polarization, near-field or far-field properties, Poynting vector, etc.). According to the disclosure, the functionality of this module **443** is now extended in such a way that the module **443** is furthermore also designed for measuring the speckle contrast. Moreover, according to the disclosure, the module **443** can also be designed for measuring the temporal duration of the light pulses.

(20) On the basis of the (sensor) signals supplied by the module **443**, according to the disclosure, the optical pulse stretcher **442** can now be detuned so as to achieve the sought reduction of the speckle contrast.

(21) FIG. 5 shows, merely schematically, a block diagram for elucidating a corresponding closed-loop control concept. In this case, in FIG. 5, the optical pulse stretcher is designated by “**510**” and the further module by “**520**”. On the basis of the sensor signals supplied by the module **520**, a closed-loop control unit **530** controls an actuator **540** and thereby brings about the aforementioned detuning of the optical pulse stretcher **510** that is suitable for reducing the speckle contrast. For example, one or more mirrors in the optical pulse stretcher **510** can be mechanically adjusted by way of the actuator **530**.

(22) The disclosure is not restricted to the above-described control or detuning of the optical pulse stretcher in order to realize the desired speckle contrast reduction. In further embodiments, on the basis of the (sensor) signals supplied by the bandwidth analysis module **441** or by the line centre analysis module **414**, the generation of the trigger signal for triggering the gas discharge can also be controlled temporally, such that e.g. the trigger signal can be generated earlier or later by a few nanoseconds (ns) or microseconds (μ s).

(23) FIG. 6 shows a set-up possible in principle of a microlithographic projection exposure apparatus **600** designed for operation in the DUV.

(24) The projection exposure apparatus **600** in accordance with FIG. 6 comprises an illumination device **610** and a projection lens **620**. The illumination device **610** serves for illuminating a structure-bearing mask (reticle) **615** with light from a light source unit **605** comprising a laser light source for example in the form of an ArF excimer laser for an operating wavelength of 193 nm (or else in the form of a KrF excimer laser for an operating wavelength of 248 nm) and also a beam shaping optical unit that generates a parallel light beam. In this case, the laser light source can be designed in the manner according to the disclosure.

(25) The illumination device **610** comprises an optical unit **611** which, inter alia, comprises a deflection mirror **612** in the example illustrated. The optical unit **611** can comprise for example a diffractive optical element (DOE) and a zoom-axicon system for producing different illumination settings (i.e. intensity distributions in a pupil plane of the illumination device **610**). Downstream of the optical unit **611** in the light propagation direction there are situated in the beam path a light mixing device (not illustrated), which in a manner known per se, for example, can comprise an arrangement of micro-optical elements suitable for achieving light mixing, and also a lens element group **613**, downstream of which is situated a field plane with a reticle masking system, which is imaged by a lens **614** disposed downstream in the light propagation direction onto the structure-bearing mask (reticle) **615** arranged in a further field plane and thereby delimits the illuminated region on the reticle. The structure-bearing mask **615** is imaged by the projection lens **620** onto a substrate, or a wafer **630**, provided with a light-sensitive layer (photoresist). In particular, the projection lens **620** can be designed for immersion operation, in which case an immersion medium is situated upstream of the wafer, or the light-sensitive layer thereof, in relation to the light propagation direction. Furthermore, it can have for example a numerical aperture NA greater than 0.85, in particular greater than 1.1.

(26) Even though the disclosure has also been described on the basis of specific embodiments,

numerous variations and alternative embodiments will be apparent to a person skilled in the art, e.g. through combination and/or exchange of features of individual embodiments. Accordingly, it goes out saying for a person skilled in the art that such variations and alternative embodiments are concomitantly encompassed by the present disclosure, and the scope of the disclosure is restricted only within the meaning of the appended patent claims and the equivalents thereof.

Claims

1. An optical system, comprising: a laser light source configured to generate a multiplicity of light pulses; a first measuring unit configured to measure a variable that is characteristic of a speckle contrast of the light generated by the laser light source; a second measuring unit configured to measure a variable that is characteristic of a bandwidth of the light generated by the laser light source; and a control unit configured to control the laser light source so that, for a light pulse sequence generated by the laser light source, a time period between respectively successive light pulses varies across the light pulse sequence depending on output signals of the first measuring unit and the second measuring unit.
2. The optical system of claim 1, further comprising an actuator, wherein the laser light source comprises an optical component, and the control unit is configured to manipulate a position of the actuator to manipulate a position of the optical component.
3. The optical system of claim 1, further comprising an optical pulse stretcher comprising a plurality of mirrors.
4. The optical system of claim 3, wherein at least one the mirrors is an optical component of the laser light source.
5. The optical system of claim 4, further comprising an actuator, wherein the control unit is configured to manipulate a position of the actuator to manipulate a position of the optical component.
6. The optical system of claim 1, wherein the laser light source comprises a laser medium, and the control unit is configured to variably adjust a temporal delay of a trigger signal generated to trigger an energy feed into the laser medium.
7. The optical system of claim 1, wherein the control unit comprises a random number generator configured to randomly vary the time period between each two successive light pulses.
8. The optical system of claim 1, wherein the control unit is configured to continuously change the time period between each two successive light pulses.
9. The optical system of claim 8, wherein the time period is continuously changed between a predefined upper value and a predefined lower value.
10. The optical system of claim 1, wherein the laser light source is configured to generate the light pulses with a repetition rate of at least 7 kHz.
11. The optical system of claim 1, wherein the optical system is configured to operate at an operating wavelength of less than 250 nm.
12. An apparatus, comprising: an optical system according to claim 1; an illumination device; and a projection lens, wherein the apparatus is a microlithographic projection exposure apparatus.
13. The optical system of claim 1, wherein the first measuring unit is configured to measure the speckle contrast of the light generated by the laser light source.
14. A method of operating an optical system comprising a laser light source configured to generate a multiplicity of light pulses and a controller configured so that, for a light pulse sequence generated by the laser light source, a time period between respectively successive light pulses varies across the light pulse sequence, the method comprising: varying the time period based on: measurement of a variable that is characteristic of a speckle contrast of the light generated by the laser light source; and a measurement of a respective bandwidth for at least one light pulse generated by the laser light source.

15. The method of claim 14, wherein the variation is adjusted to reduce an average bandwidth of the light pulses generated by the laser light source.
 16. The method of claim 14, wherein the variation is adjusted to reduce an average bandwidth difference between successive light pulses.
 17. The method of claim 14, further comprising manipulating the position of an optical component of the laser light source to at least partly effect the variation.
 18. The method of claim 14, further comprising adjusting a temporal delay of a trigger signal generated for triggering an energy feed into a laser medium of the laser light source to at least partly effect the variation.
 19. The method of claim 14, comprising varying the time period based on a measurement of the speckle contrast of the light generated by the laser light source.
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